Appendix A - full result tables

Every table below is generated from `results/raw/` by `python -m tsfdi report-figures`; the report body quotes the compact forms and points here for the per-site and per-magnitude detail.

A.1 Detection by attack family and magnitude (all cells)

var.	family	σ	n	det. rate	miss rate	delay med.	q1	q3	AUPRC	units	gain
PRCP	noise_shaped	0.5	60	0.55	0.45	39	13	95	0.048	0.247	nan
PRCP	noise_shaped	1	60	0.583	0.417	37	13.5	95	0.061	0.495	nan
PRCP	noise_shaped	1.5	60	0.717	0.283	34	12	90	0.069	0.742	nan
PRCP	noise_shaped	2	60	0.767	0.233	23.5	11.5	79	0.082	0.99	nan
PRCP	noise_shaped	3	60	0.817	0.183	19	11	70	0.137	1.563	nan
PRCP	ramp	0.5	60	0.633	0.367	75.5	24.75	112	0.076	0.215	nan
PRCP	ramp	1	60	0.633	0.367	81	28.5	102.5	0.08	0.426	nan
PRCP	ramp	1.5	60	0.7	0.3	67	28	98	0.138	0.616	nan
PRCP	ramp	2	60	0.8	0.2	74.5	28	99	0.237	0.787	nan
PRCP	ramp	3	60	0.85	0.15	61	29	96.5	0.269	1.124	nan
PRCP	scaling	0.5	60	0.683	0.317	39	14	102	0.062	0.804	0.545
PRCP	scaling	1	60	0.75	0.25	25	9	73	0.117	1.702	1.249
PRCP	scaling	1.5	60	0.833	0.167	17	9.25	54.5	0.218	2.696	2.111
PRCP	scaling	2	60	0.883	0.117	16	8	54	0.338	3.815	3.132
PRCP	scaling	3	60	0.9	0.1	13	6	44	0.394	6.407	5.647
PRCP	step	0.5	60	0.467	0.533	60.5	13	95	0.059	0.317	nan
PRCP	step	1	60	0.617	0.383	52	14	84	0.067	0.601	nan
PRCP	step	1.5	60	0.767	0.233	51	12	80.75	0.213	0.847	nan
PRCP	step	2	60	0.8	0.2	24	10.25	59	0.277	1.062	nan
PRCP	step	3	60	0.867	0.133	18	6.75	48.5	0.368	1.548	nan
TMAX	noise_shaped	0.5	160	0.538	0.462	47.5	20.25	82.5	0.041	0.452	nan
TMAX	noise_shaped	1	160	0.744	0.256	28	16	62	0.051	0.904	nan
TMAX	noise_shaped	1.5	160	0.925	0.075	21.5	12.75	40	0.131	1.356	nan
TMAX	noise_shaped	2	160	0.956	0.044	15	9	28	0.244	1.808	nan
TMAX	noise_shaped	3	160	1	0	11	5	16.25	0.454	2.712	nan
TMAX	ramp	0.5	160	0.644	0.356	64	27.5	115.5	0.053	0.59	nan
TMAX	ramp	1	160	0.794	0.206	65	39	103	0.068	1.179	nan
TMAX	ramp	1.5	160	0.894	0.106	65	40	91.5	0.203	1.769	nan
TMAX	ramp	2	160	0.956	0.044	58	39	81	0.309	2.359	nan
TMAX	ramp	3	160	0.994	0.006	48	31.5	66.5	0.344	3.538	nan
TMAX	scaling	0.5	160	0.612	0.387	28	14	63	0.048	0.59	0.045
TMAX	scaling	1	160	0.806	0.194	20	10	51	0.163	1.179	0.089
TMAX	scaling	1.5	160	0.95	0.05	13	5	29.5	0.498	1.769	0.134
TMAX	scaling	2	160	0.994	0.006	10	4	21	0.54	2.359	0.179
TMAX	scaling	3	160	0.988	0.012	5	2	12.75	0.539	3.538	0.268
TMAX	step	0.5	160	0.606	0.394	33	16	71	0.045	0.59	nan
TMAX	step	1	160	0.788	0.212	24.5	13	62.25	0.132	1.179	nan
TMAX	step	1.5	160	0.9	0.1	13	7	32	0.469	1.769	nan
TMAX	step	2	160	0.981	0.019	11	5	20	0.527	2.359	nan
TMAX	step	3	160	0.988	0.012	5	2.25	11	0.539	3.538	nan
TMIN	noise_shaped	0.5	60	0.433	0.567	32	19.75	69.75	0.042	0.629	nan
TMIN	noise_shaped	1	60	0.683	0.317	28	14	77	0.059	1.258	nan
TMIN	noise_shaped	1.5	60	0.917	0.083	18	10	52.5	0.145	1.886	nan
TMIN	noise_shaped	2	60	0.983	0.017	13	8	30.5	0.336	2.515	nan
TMIN	noise_shaped	3	60	1	0	9	5	14	0.594	3.773	nan
TMIN	ramp	0.5	60	0.517	0.483	77	37.5	101	0.052	0.744	nan
TMIN	ramp	1	60	0.767	0.233	82	49	104	0.088	1.488	nan
TMIN	ramp	1.5	60	0.933	0.067	76.5	49	94.25	0.254	2.232	nan
TMIN	ramp	2	60	0.967	0.033	67.5	49	82.5	0.329	2.976	nan
TMIN	ramp	3	60	1	0	53.5	46	63.25	0.343	4.464	nan
TMIN	scaling	0.5	60	0.533	0.467	35.5	18	77	0.045	0.744	0.126
TMIN	scaling	1	60	0.967	0.033	26.5	12	66	0.303	1.488	0.253
TMIN	scaling	1.5	60	0.967	0.033	13	4	32.75	0.538	2.232	0.379
TMIN	scaling	2	60	1	0	8	1.75	30	0.573	2.976	0.506
TMIN	scaling	3	60	1	0	4	0	16	0.607	4.464	0.758

var.	family	σ	n	det. rate	miss rate	delay med.	q1	q3	AUPRC	units	gain
TMIN	step	0.5	60	0.483	0.517	46	29	76	0.042	0.744	nan
TMIN	step	1	60	0.883	0.117	39	27	70	0.143	1.488	nan
TMIN	step	1.5	60	0.967	0.033	22	13	36.5	0.487	2.232	nan
TMIN	step	2	60	1	0	15.5	8.75	22.25	0.519	2.976	nan
TMIN	step	3	60	1	0	6	2	10.25	0.515	4.464	nan

A.2 Replay detection by window length

var.	length	n	det. rate	miss rate	delay med.	q1	q3	AUPRC
PRCP	3	60	0.267	0.733	9.5	2	19.25	0.002
PRCP	5	60	0.4	0.6	4.5	3	12.5	0.004
PRCP	7	60	0.433	0.567	5	3	10.75	0.007
PRCP	14	60	0.75	0.25	9	5	13	0.017
PRCP	21	60	0.783	0.217	9	5.5	13	0.032
TMAX	3	160	0.675	0.325	1	0	2	0.028
TMAX	5	160	0.712	0.288	1	0	3	0.146
TMAX	7	160	0.888	0.112	2	0	6	0.166
TMAX	14	160	0.962	0.037	2	0	6	0.194
TMAX	21	160	0.975	0.025	2	0	6	0.233
TMIN	3	60	0.467	0.533	1	0	2	0.005
TMIN	5	60	0.583	0.417	2	0	4	0.014
TMIN	7	60	0.767	0.233	3	0	6	0.091
TMIN	14	60	0.833	0.167	3	0	6	0.15
TMIN	21	60	0.917	0.083	4	0.5	6	0.167

A.3 False alarms per (site, variable) on clean data

site	var.	years	events	alarms/yr	in extreme	extreme d	calib. rate	target	over
CARTHAGE	TMAX	5.002	8	1.599	0	80	2	1	0.599
CARTHAGE	TMIN	5.002	3	0.6	0	27	1	1	-0.4
CARTHAGE	PRCP	5.002	4	0.8	0	16	4	1	-0.2
JENDOUBA	TMAX	5.002	12	2.399	1	92	0	1	1.399
JENDOUBA	TMIN	5.002	18	3.599	0	34	2	1	2.599
JENDOUBA	PRCP	5.002	16	3.199	0	14	4	1	2.199
GAFSA	TMAX	5.002	16	3.199	3	100	3	1	2.199
GAFSA	TMIN	5.002	2	0.4	0	50	2	1	-0.6
GAFSA	PRCP	5.002	31	6.197	0	16	3	1	5.197
MERSA_MATRUH	TMAX	5.002	19	3.798	1	92	5	1	2.798
MERSA_MATRUH	TMIN	5.002	7	1.399	0	45	1	1	0.399
KHARGA	TMAX	5.002	31	6.197	1	46	3	1	5.197
KHARGA	TMIN	5.002	27	5.398	0	63	1	1	4.398
NAVACERRADA	TMAX	5.002	7	1.399	0	73	1	1	0.399
NAVACERRADA	TMIN	5.002	5	1	0	46	1	1	-0
NAVACERRADA	PRCP	5.002	7	1.399	1	22	1	1	0.399
MONT_AIGOUAL	TMAX	5.002	5	1	0	88	5	1	-0
MONT_AIGOUAL	TMIN	5.002	24	4.798	0	42	6	1	3.798
MONT_AIGOUAL	PRCP	5.002	8	1.599	1	10	0	1	0.599
FELDBERG	TMAX	5.002	4	0.8	0	55	2	1	-0.2
FELDBERG	TMIN	5.002	10	1.999	0	52	2	1	0.999
FELDBERG	PRCP	5.002	6	1.2	0	15	0	1	0.2
CABAUW	TMAX	5.002	4	0.8	0	60	0	1	-0.2
CABAUW	TMIN	5.002	14	2.799	2	46	2	1	1.799
CABAUW	PRCP	5.002	8	1.599	0	18	1	1	0.599
HELSINKI_KAISANIEMI	TMAX	5.002	14	2.799	0	84	3	1	1.799
HELSINKI_KAISANIEMI	TMIN	5.002	8	1.599	1	70	3	1	0.599
HELSINKI_KAISANIEMI	PRCP	5.002	4	0.8	0	21	0	1	-0.2
ALTA_LUFTHAVN	TMAX	5.002	3	0.6	0	77	0	1	-0.4
ALTA_LUFTHAVN	TMIN	5.002	7	1.399	0	58	2	1	0.399
ALTA_LUFTHAVN	PRCP	5.002	9	1.799	0	15	1	1	0.799
MIAMI_INTL	TMAX	5.002	19	3.798	1	55	7	1	2.798
MIAMI_INTL	TMIN	5.002	13	2.599	0	48	4	1	1.599
MIAMI_INTL	PRCP	5.002	14	2.799	1	21	1	1	1.799

A.4 Diagnosis confidence calibration

band	n	conf.	acc.	gap
high	73	0.833	0.959	-0.126
low	88	0.543	0.25	0.293
medium	73	0.686	0.575	0.111

A.5 Efficiency: latency and memory per station

site	var.	readings	ms/read	peak MB	RSS MB	L0 μs	L2 μs	L3 μs	SP3 μs
CARTHAGE	TMAX	3653	10.499	321.23	504.652	0.26	0.495	0.015	0.016
GAFSA	TMAX	3653	20.876	321.314	510.285	0.309	0.548	0.014	0.018
NAVACERRADA	TMAX	3653	9.573	321.23	516.812	0.379	0.544	0.014	0.02
FELDBERG	TMAX	3653	7.941	321.273	520.234	0.394	0.572	0.015	0.021
HELSINKI_KAISANIEMI	TMAX	3653	19.062	321.319	520.234	0.393	0.559	0.018	0.02
MIAMI_INTL	TMAX	3653	27.795	321.365	520.234	0.385	0.549	0.025	0.024

A.6 Ablation, per variant and magnitude

variant	σ	n	hits	delay med.	q1	q3	alarms/yr	build s	det. rate	miss rate
full	1	24	16	51	12	56	2.299	44.433	0.667	0.333
full	1.5	24	22	33.75	9.625	40.375	2.299	44.433	0.917	0.083
full	2	24	23	16	8	27.625	2.299	44.433	0.958	0.042
full	3	24	23	6	4.375	7.25	2.299	44.433	0.958	0.042
gbm_expectation	1	24	21	39.25	25.625	51.375	5.298	44.966	0.875	0.125
gbm_expectation	1.5	24	22	19.75	13.25	35.5	5.298	44.966	0.917	0.083
gbm_expectation	2	24	23	7.75	6.625	16.5	5.298	44.966	0.958	0.042
gbm_expectation	3	24	23	6.25	4.875	9.75	5.298	44.966	0.958	0.042
no_fingerprint	1	24	17	47	19.75	54.75	1.599	28.322	0.708	0.292
no_fingerprint	1.5	24	20	28	9.125	40.125	1.599	28.322	0.833	0.167
no_fingerprint	2	24	23	16	8	23.125	1.599	28.322	0.958	0.042
no_fingerprint	3	24	23	6	4.375	7.25	1.599	28.322	0.958	0.042
no_reference	1	24	20	32.5	29	45.5	2.199	44.706	0.833	0.167
no_reference	1.5	24	23	30.25	23	41.125	2.199	44.706	0.958	0.042
no_reference	2	24	23	20	6.75	27.75	2.199	44.706	0.958	0.042
no_reference	3	24	24	14	9.125	17.25	2.199	44.706	1	0
no_whitening	1	24	18	27.25	23.125	48.375	4.398	44.515	0.75	0.25
no_whitening	1.5	24	21	7.5	5.25	22.5	4.398	44.515	0.875	0.125
no_whitening	2	24	21	5.25	3.5	10.75	4.398	44.515	0.875	0.125
no_whitening	3	24	21	3.25	2.25	4.25	4.398	44.515	0.875	0.125

A.7 Baselines, per attack family and magnitude

baseline	family	σ	n	det. rate	miss rate	delay med.	q1	q3	AUPRC
B0_fixed_threshold	ramp	1	20	0.95	0.05	24	8	70	nan
B0_fixed_threshold	ramp	1.5	20	0.95	0.05	27	11	58.5	nan
B0_fixed_threshold	ramp	2	20	0.95	0.05	27	11	43.5	nan
B0_fixed_threshold	ramp	3	20	1	0	31.5	11.5	45	nan
B0_fixed_threshold	ramp	5	20	1	0	24	15	38.25	nan
B0_fixed_threshold	step	1	20	0.9	0.1	22	13.75	44.5	nan
B0_fixed_threshold	step	1.5	20	0.95	0.05	13	6	29.5	nan
B0_fixed_threshold	step	2	20	1	0	7.5	4.75	15	nan
B0_fixed_threshold	step	3	20	1	0	3	1	12.75	nan
B0_fixed_threshold	step	5	20	1	0	2.5	0	10.5	nan
B1_cusum_unwhitened	ramp	1	20	1	0	76.5	55.75	105	nan
B1_cusum_unwhitened	ramp	1.5	20	1	0	69.5	43.75	80.25	nan
B1_cusum_unwhitened	ramp	2	20	1	0	55.5	42.75	68	nan

baseline	family	σ	n	det. rate	miss rate	delay med.	q1	q3	AUPRC
B1_cusum_unwhitened	ramp	3	20	1	0	42.5	32.25	50.75	nan
B1_cusum_unwhitened	ramp	5	20	1	0	25.5	18.75	32.25	nan
B1_cusum_unwhitened	step	1	20	1	0	27	7.75	44.75	nan
B1_cusum_unwhitened	step	1.5	20	1	0	9.5	5	19	nan
B1_cusum_unwhitened	step	2	20	1	0	5	3	8	nan
B1_cusum_unwhitened	step	3	20	1	0	2.5	1.75	4	nan
B1_cusum_unwhitened	step	5	20	1	0	1	1	1.25	nan
B2_ruptures_pelt	ramp	1	20	0.65	0.35	42	27	62	nan
B2_ruptures_pelt	ramp	1.5	20	0.95	0.05	42	35	62.5	nan
B2_ruptures_pelt	ramp	2	20	1	0	46.5	37	62.25	nan
B2_ruptures_pelt	ramp	3	20	1	0	40	30	57	nan
B2_ruptures_pelt	ramp	5	20	1	0	26.5	20	33	nan
B2_ruptures_pelt	step	1	20	0.6	0.4	12.5	2	88.5	nan
B2_ruptures_pelt	step	1.5	20	1	0	3	1	22	nan
B2_ruptures_pelt	step	2	20	1	0	2	1	88.5	nan
B2_ruptures_pelt	step	3	20	1	0	2	1	119	nan
B2_ruptures_pelt	step	5	20	1	0	2	1	119	nan
B3_isolation_forest[sklearn]	ramp	1	20	0.25	0.75	58	52	66	nan
B3_isolation_forest[sklearn]	ramp	1.5	20	0.3	0.7	70	55.5	88.25	nan
B3_isolation_forest[sklearn]	ramp	2	20	0.3	0.7	70	51	88.25	nan
B3_isolation_forest[sklearn]	ramp	3	20	0.6	0.4	94.5	72	107.75	nan
B3_isolation_forest[sklearn]	ramp	5	20	0.9	0.1	88	68	96	nan
B3_isolation_forest[sklearn]	step	1	20	0.2	0.8	45	30.25	60	nan
B3_isolation_forest[sklearn]	step	1.5	20	0.35	0.65	21	15	60.5	nan
B3_isolation_forest[sklearn]	step	2	20	0.45	0.55	21	15	55	nan
B3_isolation_forest[sklearn]	step	3	20	0.75	0.25	19	9.5	48	nan
B3_isolation_forest[sklearn]	step	5	20	1	0	8	1.75	15.5	nan
PROPOSED	ramp	1	20	0.7	0.3	90	65	116	0.066
PROPOSED	ramp	1.5	20	0.8	0.2	70.5	58.75	83	0.255
PROPOSED	ramp	2	20	0.95	0.05	68	58.5	82.5	0.27
PROPOSED	ramp	3	20	1	0	50.5	42.5	66.5	0.309
PROPOSED	ramp	5	20	1	0	36	30.75	50.75	0.302
PROPOSED	step	1	20	0.75	0.25	40	25	48	0.22
PROPOSED	step	1.5	20	0.9	0.1	22	12.5	27	0.446
PROPOSED	step	2	20	1	0	10	6.5	20	0.475
PROPOSED	step	3	20	1	0	6	4	8	0.51
PROPOSED	step	5	20	1	0	2	1	3	0.372

A.8 Baseline false alarms

baseline	sites	FA med.	FA max.
B3_isolation_forest[sklearn]	4	0.3	0.6
B2_ruptures_pelt	4	0.8	1.399
PROPOSED	4	2.199	3.199
B0_fixed_threshold	4	6.497	8.197
B1_cusum_unwhitened	4	8218.12	12783.8

A.9 The proposed suite at three operating points

budget	n	det. rate	delay med.	FA med.
0.1	160	0.812	51.25	1.1
0.5	160	0.862	47.5	1.499
1	160	0.888	47.75	2.199

A.10 Cross-site generalisation

site	n	det. rate	miss rate	delay med.	q1	q3	AUPRC
EG000062306	160	0.975	0.025	21	9.75	53	0.32
FI000000304	160	0.975	0.025	12	5	28	0.352
GME00120934	160	0.906	0.094	23	12	64	0.325
SP000008215	160	1	0	11	2	40	0.52
TS000060725	160	1	0	13	4	37.5	0.415
TSM00060715	160	1	0	17.5	6	44	0.424
TSM00060745	160	1	0	10	5	15	0.511
USW00012839	160	1	0	8	3	20	0.559

A.11 Evasion: per site and level

site	var.	level	detected	day	detectors	delay	cum. σ	mean σ	days	s	noise	safety	sign	incr. σ	achiev. σ
CARTHAGE	TMAX	A1a	True	1868	D3_replay_ncc	D6_fingerprint	42	49.403	nan	180	1.327	none	nan	nan	nan
CARTHAGE	TMAX	A1b	True	1959	D3_replay_ncc	42	80.152	nan	180	1.239	white	nan	nan	nan	nan
CARTHAGE	TMAX	A1c	True	1959	D3_replay_ncc	42	76.985	nan	180	1.195	ar1	nan	nan	nan	nan
CARTHAGE	TMAX	A2	False	nan	nan	nan	37.5	1.21	31	2.056	nan	0.15	-1	0.156	nan
CARTHAGE	TMAX	A2-blackbox	False	nan	nan	nan	74.473	nan	240	7.205	nan	nan	nan	nan	0.908
GAFSA	TMAX	A1a	True	1861	D6_fingerprint	35	101.45	nan	180	1.201	none	nan	nan	nan	nan
GAFSA	TMAX	A1b	True	1952	D3_replay_ncc	D6_fingerprint	168	140.995	nan	180	1.283	white	nan	nan	nan
GAFSA	TMAX	A1c	True	1980	D3_replay_ncc	179	135.767	nan	180	1.24	ar1	nan	nan	nan	nan
GAFSA	TMAX	A2	True	2085	nan	nan	14.1	1.41	10	2.203	nan	0.15	-1	0.059	nan
GAFSA	TMAX	A2-blackbox	False	nan	nan	253	87.949	nan	240	7.508	nan	nan	nan	nan	0.557
NAVACERRADA	TMAX	A1a	True	1829	D6_fingerprint	56	206.165	1.145	180	1.355	none	nan	nan	nan	nan
NAVACERRADA	TMAX	A1b	True	1829	D4_spike	D6_fingerprint	154	279.485	1.553	180	1.349	white	nan	nan	nan
NAVACERRADA	TMAX	A1c	True	1829	D1_window_shift_7	D3_block_contrast	D3_replay_ncc	D6_fingerprint	9	267.42	1.486	180	1.253	ar1	nan
NAVACERRADA	TMAX	A2	True	1910	D3_replay_ncc	D6_fingerprint	91	337.4	1.599	211	2.714	nan	0.15	-1	1.406
NAVACERRADA	TMAX	A2-blackbox	False	nan	D3_replay_ncc	D6_fingerprint	3	12.656	0.053	240	7.907	nan	nan	nan	nan
FELDBERG	TMAX	A1a	True	1875	D6_fingerprint	56	165.833	0.921	180	1.318	none	nan	nan	nan	nan
FELDBERG	TMAX	A1b	True	1882	D6_fingerprint	56	250.06	1.389	180	1.347	white	nan	nan	nan	nan
FELDBERG	TMAX	A1c	True	1882	D6_fingerprint	84	238.511	1.325	180	1.299	ar1	nan	nan	nan	nan
FELDBERG	TMAX	A2	True	1896	D3_replay_ncc	D6_fingerprint	70	301.9	1.348	224	2.817	nan	0.15	-1	1.258
FELDBERG	TMAX	A2-blackbox	False	nan	D6_fingerprint	56	177.188	0.738	240	7.624	nan	nan	nan	nan	0.738
HELSINKI_KAISANIEMI	TMAX	A1a	True	1882	D6_fingerprint	56	155.636	0.865	180	1.262	none	nan	nan	nan	nan
HELSINKI_KAISANIEMI	TMAX	A1b	True	1910	D6_fingerprint	84	240.745	1.337	180	1.358	white	nan	nan	nan	nan
HELSINKI_KAISANIEMI	TMAX	A1c	True	1910	D6_fingerprint	84	231.472	1.286	180	1.268	ar1	nan	nan	nan	nan
HELSINKI_KAISANIEMI	TMAX	A2	True	1966	D6_fingerprint	84	111.7	1.269	88	3.181	nan	0.15	-1	0.465	nan
HELSINKI_KAISANIEMI	TMAX	A2-blackbox	False	nan	D6_fingerprint	84	19.687	0.082	240	7.722	nan	nan	nan	nan	0.082
MIAMI_INTL	TMAX	A1a	True	1861	D6_fingerprint	35	147.215	0.818	180	1.243	none	nan	nan	nan	nan
MIAMI_INTL	TMAX	A1b	True	1902	D6_fingerprint	105	218.338	1.213	180	1.344	white	nan	nan	nan	nan
MIAMI_INTL	TMAX	A1c	True	1902	D6_fingerprint	140	216.379	1.202	180	1.324	ar1	nan	nan	nan	nan
MIAMI_INTL	TMAX	A2	True	1889	D3_replay_ncc	D6_fingerprint	45	349.9	1.62	216	2.658	nan	0.15	-1	1.458
MIAMI_INTL	TMAX	A2-blackbox	False	nan	D1_step_cusum	D1_window_shift_3	D1_window_shift_7	D2_drift_cusum	D2_slope_14	D2_slope_30	D3_block_contrast	D3_replay_ncc	D6_fingerprint	76	14.062

A.13 Calibration summary (all 34 pairs)

site	var.	σ res.	σz	range	σ/range	AR	rank-G	FA test	FA calib.
CARTHAGE	TMAX	1.106	0.999	18.709	0.059	2	False	1.599	0.4
CARTHAGE	TMIN	1.422	0.996	16.015	0.089	1	False	0.6	0.2
CARTHAGE	PRCP	0.186	0.986	0.001	135	1	True	0.8	0.8
JENDOUBA	TMAX	1.253	0.994	22.253	0.056	2	False	2.399	0
JENDOUBA	TMIN	1.761	0.97	16.1	0.109	1	False	3.599	0.4
JENDOUBA	PRCP	0.201	1.033	0	4262.66	5	True	3.199	0.8
GAFSA	TMAX	1.269	1.002	23.123	0.055	5	False	3.199	0.6
GAFSA	TMIN	1.554	1.005	20.191	0.077	1	False	0.4	0.4
GAFSA	PRCP	0.001	1.109	0	3303.36	3	True	6.197	0.6
MERSA_MATRUH	TMAX	0.994	0.999	12.906	0.077	5	False	3.798	1
MERSA_MATRUH	TMIN	1.397	1.041	14.324	0.098	2	False	1.399	0.2
KHARGA	TMAX	0.677	0.995	20.296	0.033	3	False	6.197	0.6
KHARGA	TMIN	2.639	1.03	19.415	0.136	1	False	5.398	0.2
NAVACERRADA	TMAX	1.574	1.003	21.957	0.072	4	False	1.399	0.2
NAVACERRADA	TMIN	2.762	1.006	16.481	0.168	5	False	1	0.2
NAVACERRADA	PRCP	0.193	1.469	0.793	0.244	0	True	1.399	0.2
MONT_AIGOUAL	TMAX	1.857	0.985	19.086	0.097	5	False	1	1
MONT_AIGOUAL	TMIN	2.298	1.005	15.423	0.149	3	False	4.798	1.2
MONT_AIGOUAL	PRCP	0.365	0.986	0.633	0.576	0	True	1.599	0
FELDBERG	TMAX	1.856	1.001	17.693	0.105	5	False	0.8	0.4
FELDBERG	TMIN	2.512	0.99	15.984	0.157	3	False	1.999	0.4
FELDBERG	PRCP	0.586	1.055	0.698	0.84	4	True	1.2	0
CABAUW	TMAX	0.863	0.998	16.957	0.051	4	False	0.8	0
CABAUW	TMIN	1.263	0.988	13.136	0.096	1	False	2.799	0.4
CABAUW	PRCP	0.325	0.985	0.355	0.916	0	True	1.599	0.2
HELSINKI_KAISANIEMI	TMAX	0.987	1.011	21.668	0.046	2	False	2.799	0.6
HELSINKI_KAISANIEMI	TMIN	1.22	0.982	18.623	0.065	4	False	1.599	0.6
HELSINKI_KAISANIEMI	PRCP	0.436	1.013	0.35	1.244	1	True	0.8	0
ALTA_LUFTHAVN	TMAX	1.771	0.995	21.347	0.083	3	False	0.6	0

site	var.	σ res.	σz	range	σ/range	AR	rank-G	FA test	FA calib.
ALTA_LUFTHAVN	TMIN	2.044	1.024	22.183	0.092	2	False	1.399	0.4
ALTA_LUFTHAVN	PRCP	0.437	0.979	0.291	1.503	1	True	1.799	0.2
MIAMI_INTL	TMAX	0.912	1.006	7.123	0.128	5	False	3.798	1.4
MIAMI_INTL	TMIN	1.216	1.004	7.694	0.158	3	False	2.599	0.8
MIAMI_INTL	PRCP	0.705	0.978	0.928	0.76	0	True	2.799	0.2

A.14 Excluded pairs

None: all 34 design pairs are modelled on this bundle.

A.15 Figures referenced from the body



Figure 3 — spectral-fingerprint example: the self-normalised distance moves when the sensor is replaced by a copy of the reference.

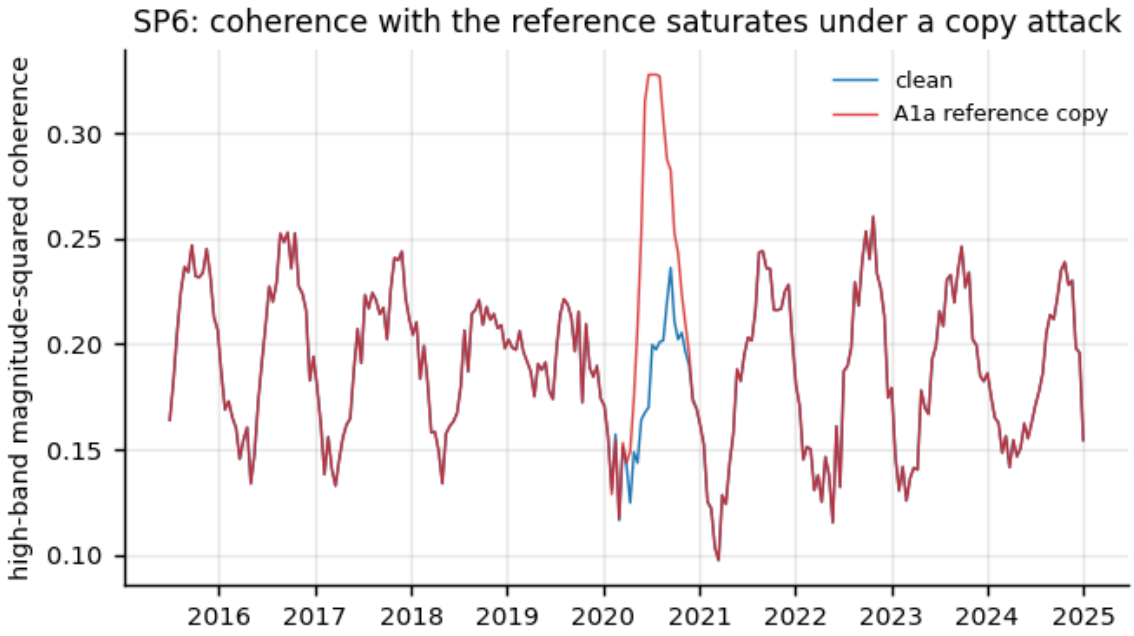


Figure 4 — sensor/reference magnitude-squared coherence per band, honest versus attacked.